VISIONS CLASSES

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MAHARASHTRA STATE BOARD OF SECONDARY & HIGHER SECONDARY EDUCATION

SECONDARY SCHOOL CERTIFICATE (SSC) EXAMINATION

Mathematics (Part – I) | Standard X

Chapter 3 : Arithmetic Progression (Practice Set 3.1 to 3.3)

Total Marks : **40** Time : **1 Hour 15 Minutes**

General Instructions :

1. All questions are **compulsory**.
2. Use of calculator is **not allowed**.
3. Show all steps clearly for full marks. Figures to the right indicate marks.
4. Draw neat diagrams wherever necessary.
5. For Q.3, attempt any **3 out of 4**. For Q.4, attempt any **2 out of 3**.

Q. 1 Choose the Correct Alternative (MCQ) **4 Marks (1 each)**

(1) Which of the following sequences is an Arithmetic Progression?

(A) 1, 3, 6, 10, . . . (B) 2, 4, 6, 8, . . .

(C) 1, 4, 9, 16, . . . (D) 2, 6, 18, 54, . . .

(2) The first term of an A.P. is 8 and common difference is −5. What is the 4th term?

(A) 23 (B) 3

(C) −7 (D) −2

(3) If $t_n = 5n - 2$, then the 30th term of the A.P. is :

(A) 150 (B) 148

(C) 152 (D) 146

(4) The sum of first 10 natural numbers is :

(A) 45 (B) 100

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(C) 50 (D) 55

Q. 2 Complete the Following Activities **8 Marks**

(A) Write the correct number in the given boxes from the following A.P. : *[4 Marks]*

1, 8, 15, 22, . . .

Here $a =$ _____ , $t_1 =$ _____ , $t_2 =$ _____ , $t_3 =$ _____

$t_2 - t_1 =$ _____ − _____ = _____

$t_3 - t_2 =$ _____ − _____ = _____

$\therefore\ d =$ _____

(B) Complete the following activity to find S_{27}, given $a = 6$, $d = 3$: *[4 Marks]*

$S_n = n/2\ [$ _____ $+ (n-1)\ d]$

$S_{27} = 27/2\ [\ 12 + (27-1) \times$ _____ $]$

$= 27/2 \times$ _____

$= 27 \times 45 =$ _____

Q. 3 Solve the Following (Any 3 out of 4) **12 Marks (3 each)**

(1) Which of the following sequences are A.P.? If they are A.P., find the common difference.

(i) −10, −6, −2, 2, . . .

(ii) 0, −4, −8, −12, . . .

(iii) 0.3, 0.33, 0.333, . . .

(2) Find the first term and common difference for each of the following A.P. :

(i) 5, 1, −3, −7, . . .

(ii) 0.6, 0.9, 1.2, 1.5, . . .

(iii) 127, 135, 143, 151, . . .

(3) The 11th term and the 21st term of an A.P. are 16 and 29 respectively. Find the 41st term of that A.P.

(4) In the natural numbers from 10 to 250, how many are divisible by 4?

Q. 4 Solve the Following (Any 2 out of 3) — 12 Marks (6 each)

(1) The 10th term and the 18th term of an A.P. are 25 and 41 respectively. Find :

(i) The 38th term of the A.P.

(ii) The value of n, if the nth term is 99.

(2) Find the sum of all even numbers between 1 and 350.

Hint : Form an A.P. of even numbers, find n, then apply S_n formula.

(3) In an A.P., the 19th term is 52 and the 38th term is 128. Find the sum of the first 56 terms.

Q. 5 Problem Solving (Word Problem — Practice Set 3.2) — 4 Marks

11, 8, 5, 2, . . .

In this A.P., which term is the number **−151**?

(Show complete working using the formula $t_n = a + (n-1)d$)

Marks Distribution — For Reference

Q. No.	Type	Topic	Marks
Q. 1	MCQ (4 × 1)	Basic concepts of A.P. — Practice Set 3.1 & 3.2	4
Q. 2	Activity (A + B)	A.P. terms & Sum formula — Practice Set 3.1 & 3.3	8
Q. 3	Short Answer (Any 3)	nth term applications — Practice Set 3.1 & 3.2	12
Q. 4	Long Answer (Any 2)	Advanced A.P. & Sum problems — Practice Set 3.2 & 3.3	12
Q. 5	Word Problem	nth term — Practice Set 3.2 (Q.8)	4
Total			**40**